

Daksh — Class X Study Test

Subject: Mathematics | **Test:** Test 3 (Day 21, 30 May 2026) — Weekly
Time Allowed: 3 Hours • **Maximum Marks:** 80

Syllabus covered: Real Numbers • Polynomials • Pair of Linear Equations in Two Variables • Quadratic Equations • Arithmetic Progressions • Triangles (cycle revision)

General Instructions:

- (i) This question paper contains five sections — A, B, C, D and E.
- (ii) Section A has Multiple Choice / Very Short Answer Questions of 1 mark each.
- (iii) Section B has Short Answer Questions of 2 marks each.
- (iv) Section C has Short Answer Questions of 3 marks each.
- (v) Section D has Long Answer Questions of 5 marks each.
- (vi) Section E has Case-based / Source-based questions of 4 marks each.
- (vii) All questions are compulsory. Internal choice has been provided in some questions wherever indicated.
- (viii) Use of calculators is not permitted.

SECTION A — Multiple Choice / Very Short Answer Questions (20 × 1 = 20 marks)

Choose the correct option / answer.

- Q1.** The HCF of 144 and 198 is: (a) 12 (b) 18 (c) 36 (d) 9
- Q2.** The smallest number that is divisible by both 12 and 15 is: (a) 30 (b) 45 (c) 60 (d) 90
- Q3.** The number of zeroes of the polynomial whose graph is given as a parabola crossing the x-axis at exactly one point is: (a) 0 (b) 1 (c) 2 (d) 3
- Q4.** If α and β are zeroes of $x^2 - 4x + 3$, then $1/\alpha + 1/\beta = ?$ (a) $4/3$ (b) $3/4$ (c) 1 (d) 4
- Q5.** If the lines $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, then $k = ?$ (a) $5/4$ (b) $15/4$ (c) $3/2$ (d) $2/5$
- Q6.** The graph of the equation $y = 0$ represents: (a) y-axis (b) x-axis (c) origin (d) line parallel to x-axis
- Q7.** The equation $x^2 + 1 = 0$ has: (a) two real roots (b) one real root (c) no real roots (d) two equal roots
- Q8.** If the sum of roots of $ax^2 + bx + c = 0$ is equal to the product of roots, then: (a) $b = c$ (b) $b + c = 0$ (c) $b - c = 0$ (d) $b = 0$
- Q9.** The 17th term of the AP: 4, 9, 14, 19, ... is: (a) 84 (b) 88 (c) 80 (d) 76
- Q10.** The sum of the first n natural numbers is: (a) $n(n+1)/2$ (b) $n(n-1)/2$ (c) $n^2(n+1)/2$ (d) $n(n+1)(2n+1)/6$
- Q11.** Which of the following is an AP? (a) 1, 4, 9, 16, ... (b) 2, 4, 8, 16, ... (c) 1, 3, 5, 7, ... (d) 1, $1/2$, $1/3$, $1/4$, ...
- Q12.** If two triangles are similar, the ratio of their areas equals: (a) ratio of corresponding sides (b) square of ratio of corresponding sides (c) cube of ratio of corresponding sides (d) ratio of perimeters
- Q13.** $\triangle ABC \sim \triangle PQR$. If $AB = 4$ cm, $BC = 5$ cm, $AC = 6$ cm and $PQ = 8$ cm, then $PR = ?$ (a) 8 cm (b) 10 cm (c) 12 cm (d) 14 cm
- Q14.** A man goes 10 m due east and then 24 m due north. The distance from the starting point is: (a) 20 m (b) 26 m (c) 30 m (d) 34 m
- Q15.** If $\sin A = 1/2$ and A is acute, then $A = ?$ (a) 30° (b) 45° (c) 60° (d) 90°
- Q16.** The decimal expansion of $17/8$ is: (a) terminating (b) non-terminating recurring (c) non-terminating non-recurring (d) cannot say
- Q17.** If the discriminant of a quadratic equation is negative, what kind of roots does it have?
- Q18.** Find the number of terms in the AP: 7, 13, 19, ..., 205.
- Q19.** If $\triangle ABC \sim \triangle DEF$, $\angle A = 50^\circ$, $\angle B = 60^\circ$, find $\angle F$.
- Q20.** Write the prime factorisation of 156.

SECTION B — Short Answer-I (5 × 2 = 10 marks)

- Q21.** Show that $5 - \sqrt{3}$ is irrational, given that $\sqrt{3}$ is irrational.
- Q22.** Find the zeroes of the polynomial $x^2 + 7x + 10$ and verify the relationship between the zeroes and coefficients.

Q23. Solve: $3x - 5y = -1$ and $x - y = -1$.

Q24. Find the discriminant of the quadratic equation $3x^2 - 5x + 2 = 0$ and hence comment on the nature of its roots.

Q25. In $\triangle ABC$ right-angled at B, $AB = 24$ cm, $BC = 7$ cm. Find AC.

SECTION C — Short Answer-II ($6 \times 3 = 18$ marks)

Q26. Three bells toll together at intervals of 9, 12 and 15 minutes respectively. If they toll together at 8:00 a.m., when will they next toll together?

Q27. If the sum of zeroes of the polynomial $2x^2 - kx + 6$ is 3, find the value of k.

Q28. A father is three times as old as his son. After 12 years, he will be twice as old as his son. Find their present ages.

Q29. The sum of a number and its reciprocal is $50/7$. Find the number(s).

Q30. Find the sum of all natural numbers between 100 and 500 which are divisible by 8.

Q31. In $\triangle ABC$, $AD \perp BC$. Prove that $AB^2 - BD^2 = AC^2 - DC^2$.

SECTION D — Long Answer ($4 \times 5 = 20$ marks)

Q32. Prove that $\sqrt{7}$ is irrational.

Q33. A boat takes 4 hours to go 44 km downstream and the same time to go 28 km upstream. If the speed of the stream is the same in both directions, find the speed of the boat in still water and the speed of the stream.

Q34. The sum of the first 7 terms of an AP is 49 and that of its first 17 terms is 289. Find the sum of its first n terms.

Q35. State and prove the converse of the Basic Proportionality Theorem (Converse of BPT). Use it to prove that, in a triangle, the line joining the mid-points of two sides is parallel to the third side.

SECTION E — Case-based / Source-based ($3 \times 4 = 12$ marks)

Q36. Case Study (AP — Real-Life Application). A theatre has 30 rows of seats. The first row has 20 seats, the second has 22, the third has 24, and so on (each next row has 2 more seats than the previous one).

- (i) Form an AP for the number of seats in successive rows. (1)
- (ii) How many seats are there in the 25th row? (1)
- (iii) Find the total number of seats in the theatre. (2)

Q37. Case Study (Quadratic Equations — Real-Life Application). The hypotenuse of a right triangle is 13 cm. If one of its sides exceeds the other by 7 cm, find the sides.

- (i) If the smaller side is x cm, write the equation. (2)
- (ii) Find the value of x. (2)

Q38. Case Study (Triangles — Similarity in Real Life). A vertical pole 6 m high casts a shadow of 4 m on the ground. At the same time, a tower casts a shadow of 28 m.

- (i) Are the two triangles formed similar? Give one reason. (1)
- (ii) Find the height of the tower. (2)
- (iii) If the height of a man (1.8 m tall) standing nearby were measured, what would be the length of his shadow at the same time? (1)